RAK WisMesh 1000mW (+30dBm) 25W MPPT Solar Repeater for MeshCORE or Meshtastic (868MHz and 915MHz)

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Version 1.1

Thank you for purchasing the RAKwireless WisMesh 1W (+30dBm) 25W MPPT Solar repeater for MeshCORE or Meshtastic (EU868, US915, ANZ915).

To get the full version of this guide, please scan the the QR code in the top left corner using your phone's camera:

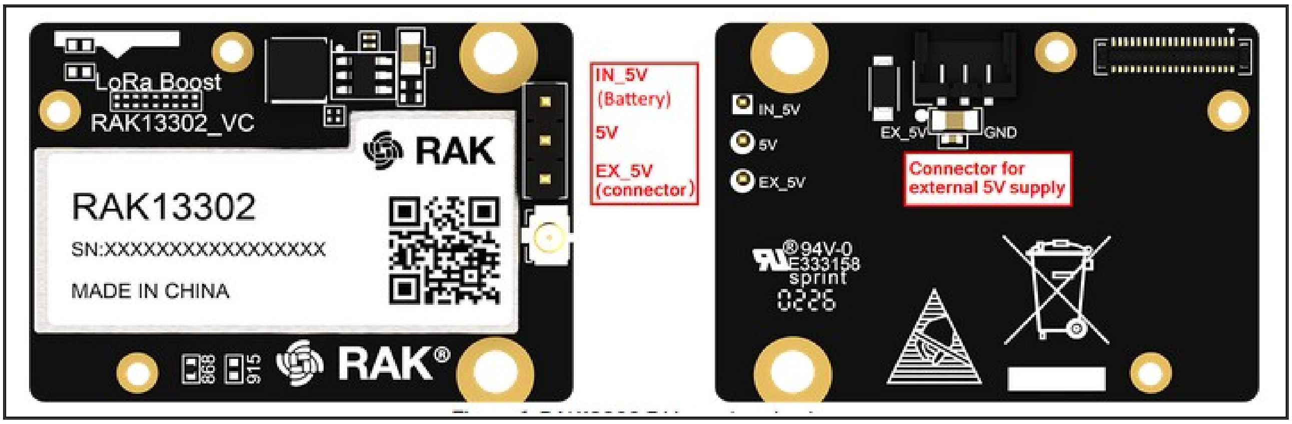
The 25W MPPT Solar MeshCORE / Meshtastic repeater uses a Single Board Computer (SBC). There are two versions, either 868MHz (EU/UK) or 915MHz (US). At the heart of the RAK WisMesh 1000mW SBC is a Nordic nRF52840 MCU integrated Bluetooth and Wi-Fi, a +22dBm Semtech SX1262 LoRa modem and SkyWorks SKY66122 +8dBm PA (Power Amplifier)

For more information about the RAKwireless WisMesh 1000mW, please scan the above QR code using your phone's camera.

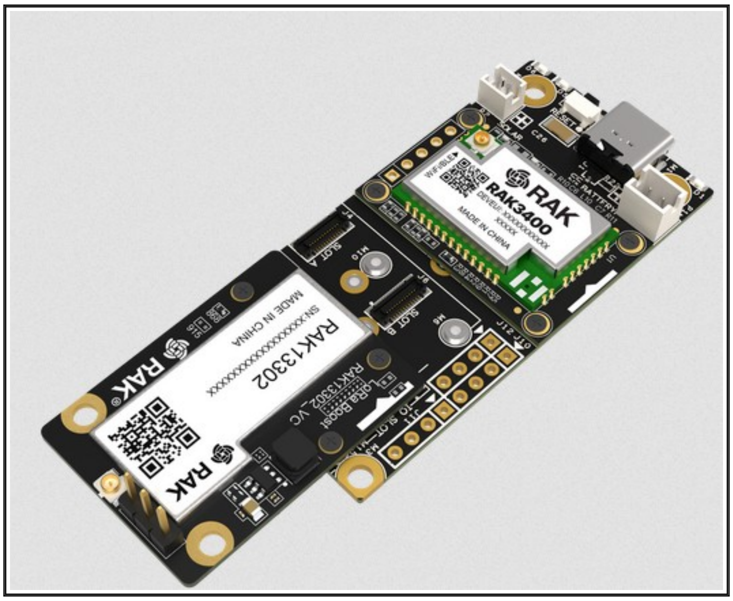
RAKireless WisMesh 1W Booster Power Supplies

The [RAK WisMesh 1W Booster Starter Kit]() requires a stable 5V supply to output full RF power of +30dBm (1000mW). The 5V supply delivered via USB-C is stepped down to 3.3V via a LDO (Low Drop Out voltage regulator) on the RAK1907 board); but sadly this is 3.3V rail is unable to deliver the surge current that the SkyWorks SKY66122 PA inside the RAK13302 demands when transmitting at +30dBm. To overcome this, the [RAK WisMesh 1W Booster Starter Ki]()t (RAK3400, RAK19007 and RAK13302) has been designed to obtain a stable auxillary 5V power supply from two external sources.

One option available is to provide a stable 5V supply directly to the RAK13302 module through it's EX_5V connector located on the back (picture below right). If this option is selected, then the three pin jumper next to the IPEX connector on the top side of the RAK13302 module needs to be placed in the "EXT_5V" position (picture below left).

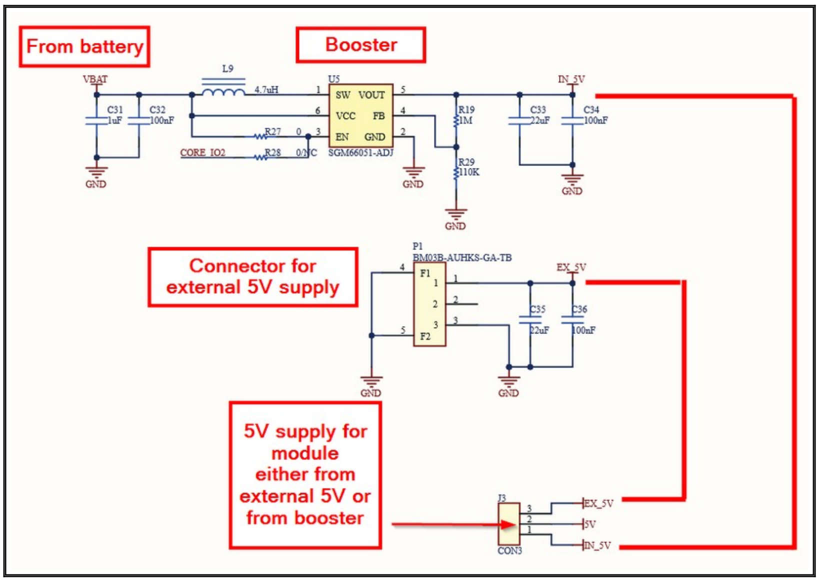


Another option is to provide the SkyWorks SKY66122 PA with a stable 5V supply that is generated from a high surge current 3.7V Lithium-Ion battery connected to the JST PH 2.0 connector on the RAK19007 board (the connector top right of the RAK3400 in the picture below).

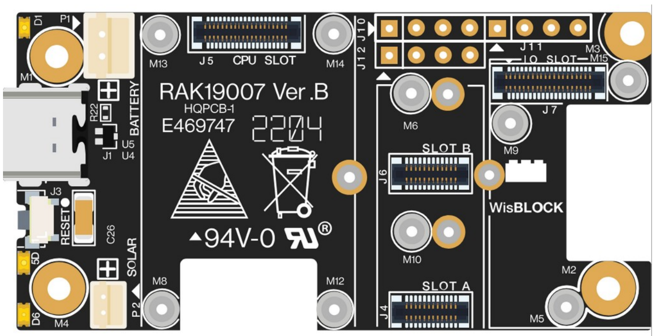


To achieve this, the RAK13302 base board uses a SGM66051 DC-DC Synchronous Booster Converter to step the supply provided by external 3.7V Lithium-Ion batteies up to 5V; this dedicated 5V supply is then used to power the SkyWorks SKY66122 PA. To select this option, the three pin jumper next to the IPEX connector (top side of the RAK13302 module) needs to be set to "IN_5V". This is the default setting when a RAK 1W board is shipped.

The block diagram below shows these two options in detail:



The "IN_5V" option is how your 25W MPPT Solar RAK WisMesh 1000mW (+30dBm) 25W MPPT Solar Repeater is configured to power the SkyWorks SKY66122 PA on the RAK13302 module. The bank of six or eight 18650 batteries in the door of your repeater are connected in parallel and supply 3.7V (or up to 4.2V when fully charger) to the SGM66051 DC-DC boost converter on rhe RAK13002 module via connector P2 (JST PH 2.0) on the RAK19007 base board. The six or eight 18650 batteries are then charged by an external MPPT controller on the host PCB. Although the RAK19007 base board does provide it's own Solar port (via P2) and Lithium-Ion battery charger (via P1):



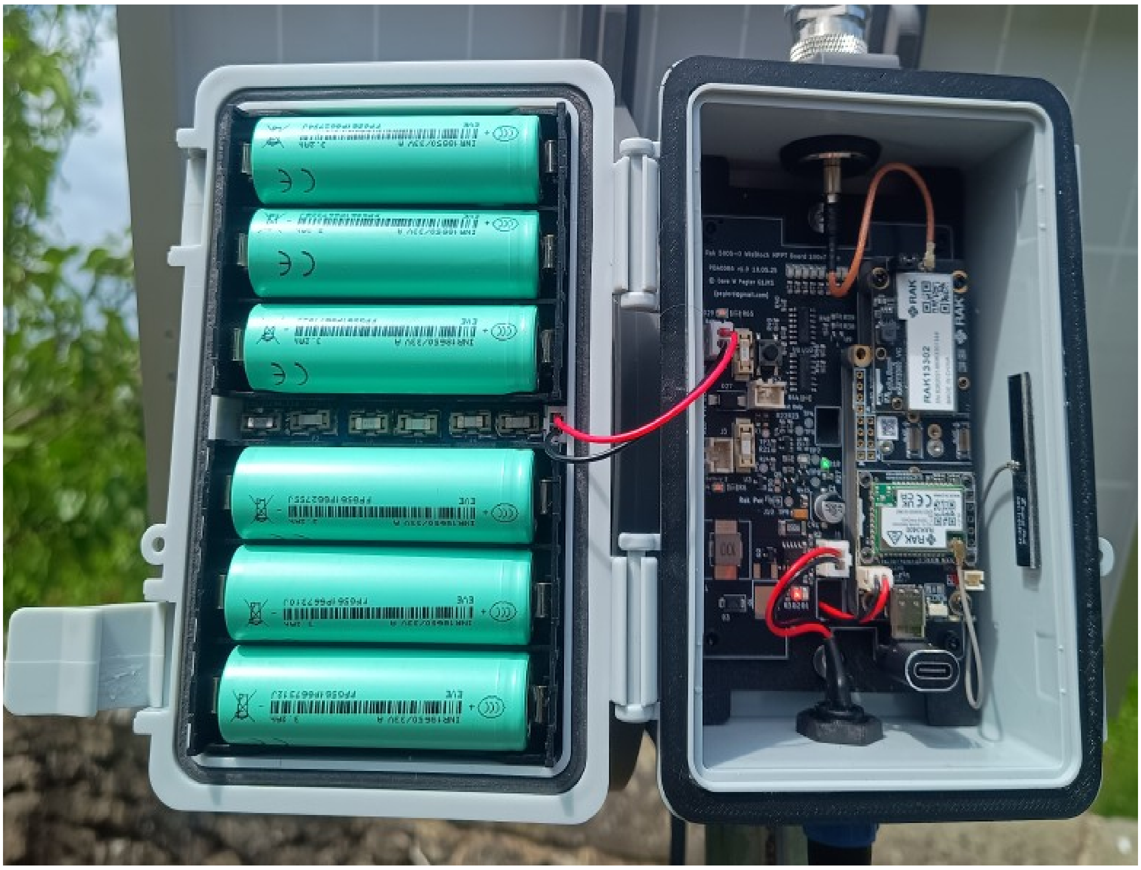
the board uses a simple TP4054 Lithium-Ion charger IC which is primarily designed to charger Lithium batteries from a constant 5V supply (i.e. USB-C), not a solar panel. The TP4054 is not a MPPT (or even a PWM) solar controller IC, and so, at best, will only work if used with low capacity 5V solar panels, and will only really work when it's fairly sunny. Using a solar panel that generates more than 5V (e.g. standard 12V solar panels with Vmp of between 18V and 20V) will destroy the TP4054 and the electronics around it:

> **WARNING**
>
> - Only 3.7-4.2 V Rechargeable Li-Ion batteries are supported. Do not use other types of batteries with the system.
> - Only 5 V solar panels are supported. Do not use 12 V solar panels. It will destroy the charging unit and eventually other electronic parts.

So that the [RAK WisMesh 1W Booster Starter Ki] can be powered from a large bank of 3.7V Lithium-Ion batteries (e.g. 18650, 21700, 26650) which will be continually charged from a 12V (i.e. 17V to 20V Vmp) solar panel (e.g. 15W, 25W, 30W and 50W), the RAK 19007 base board is mounted on a custom designed MPPT Solar controller PCB. The PCB supplies a high surge current 3.7V supply to the RAK19007 board via P2 (JST PH 2.0 connector), keeps the 18650 batteries charged up (from a 25W solar panel) and prevents the 18650 batteries from becoming over-discharged when there is a lack of light for a substantial period. The following section will now describe this MPPT Solar controller board in more detail.

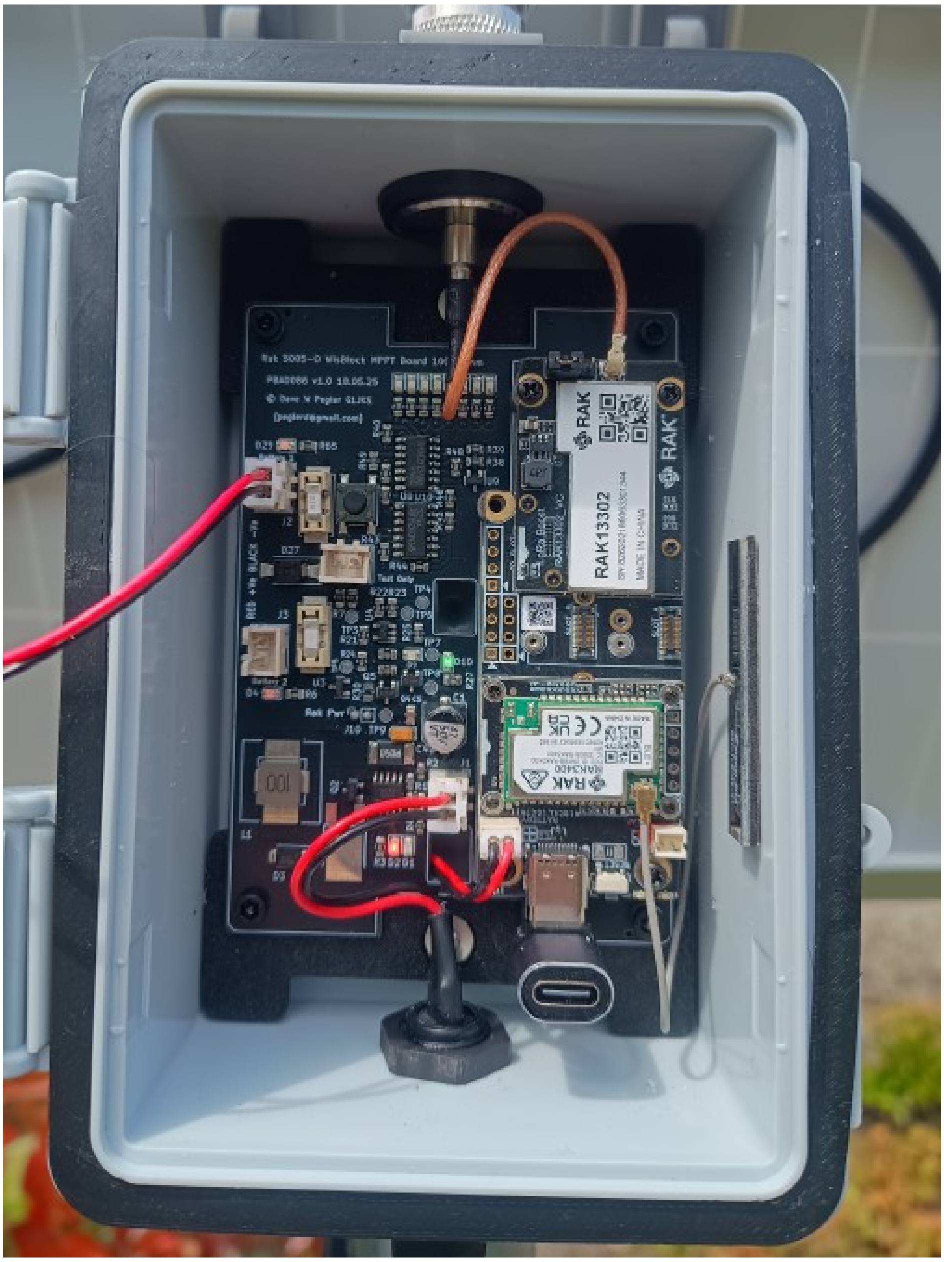
MPPT Solar Controller Overview

The RAKwireless WishMesh 1000mW (+30dBm) SBC (Single Board Computer) PCB is not designed for outdoor use, and so in order for it to operate in remote or harsh conditions it is housed in a IP67 outdoor case with a six or eight 3.7V 18650 Lithium-Ion battery compartment mounted neatly in the door. To keep the 18650 batteries charged up, a 25W Monocrystalline Photovoltaic (PV) Solar Panel is also supplied along with a MPPT charger PCB - on which the RAKwireless WishMesh 1000mW (+30dBm) SBC is mounted. Power from the batteries is supplied to the RAKwireless WishMesh 1000mW (+30dBm) SBC, via the MPPT PCB and then through a PH 2.0mm connector (RAK19007 P2), as show below.



One of the challenges with powering electronics from Solar, especially in northern hemisphere countries, is that during the winter months there are long periods of dull, cloudy, overcast days, which is not ideal for using Solar Panels to charge Lithium-Ion batteries. In the northern hemisphere the longest day (aka Summer Solstice) occurs on the 21st June, where there is sixteen hours of daylight. However on the shortest day (aka Winster Solstice), which occurs on the 21st December, there is just under eight hours of daylight.

To cope with periods of short, cloudy, overcast days, MPPT or Maximum Power Point Tracking is a technique that has been developed to adjusts the load seen by Solar Panels (i.e. input impedance) to maximise the amount of power transferred – irrespective of the amount of solar radiation and ambient temperature. MPPT controllers are particularly useful in low-light conditions, such as cloudy or overcast days, because (unlike traditional PWM controllers) they can still extract some energy from a Solar Panel. They also perform well in cold weather, where they can boost module output more than in warm weather. MPPT controllers are more complex and expensive than PWM controllers because of the additional circuitry they contain. However, the benefits of improved energy production and faster return on investment can often outweigh the initial cost.



For this reason your RAKwireless WishMesh 1000mW (+30dBm) 25W MeshCORE / Meshtastic Solar repeater contains a MPPT controller PCB which attempts to charge the six 18650 Lithium-Ion batteries whatever the weather can throw at it. Above is a photograph of the MPPT controller PCB in your RAKwireless WishMesh 1000mW (+30dBm) MeshCORE / Meshtastic 25W Solar repeater and what follows is a short description of what the various LEDs indicate. It is important to understand what the LEDs show as the most important job of the PCB is to prevent the 18650 Lithium-Ion batteries from becoming over-discharged if there is insufficient sunlight for the MPPT charger to keep them topped up. If Lithium-Ion batteries are allowed to discharge below 3.1V, they will become permanently damaged which will cause capacity degradation.

On the MPPT controller PCB (pictured overleaf), you will observer that there are two JST 2.54mm (XH) battery sockets labelled “Battery 1” and “Battery 2” – with fuses next to them. The Red LED above the “Battery 1” socket indicates that the MPPT controller PCB is receiving power from the battery board. If the Red LED above the “Battery 1” is on but no other LEDs are illuminated (See D9 and D10 below) this would indicate that the Fuse next to “Battery 1” has blow and needs replacing. Please see section **“18650 Lithium-Ion Batteries and Fuses”** below for more information.

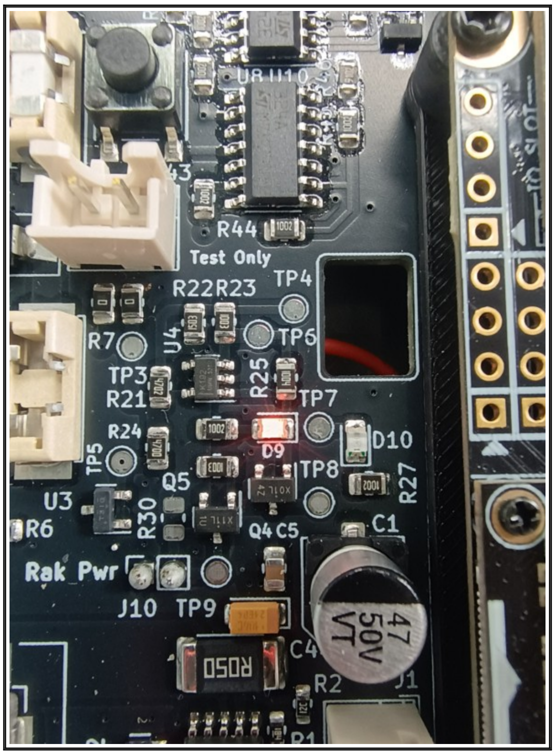
The Red LED (D2) bottom middle, indicates that the MPPT controller is extracting power from the Solar Panel and it is charging the Lithium-Ion batteries. The brightness of this LED indicates how much power it is extracting from the Solar Panel. On a dull day it will be quite dim, on a bright sunny day it will be very bright. Next to the Red LED (D2) is a Green LED (D1) which the MPPT controller illuminates when the 18650 Lithium-Ion batteries have reached their maximum charge voltage (approx 4.2v), and it has stop charging them to prevent overcharging. It is very unlikely you will see this LED illuminated unless the solar panel is seeing a lot of sunlight for a extended period of time.

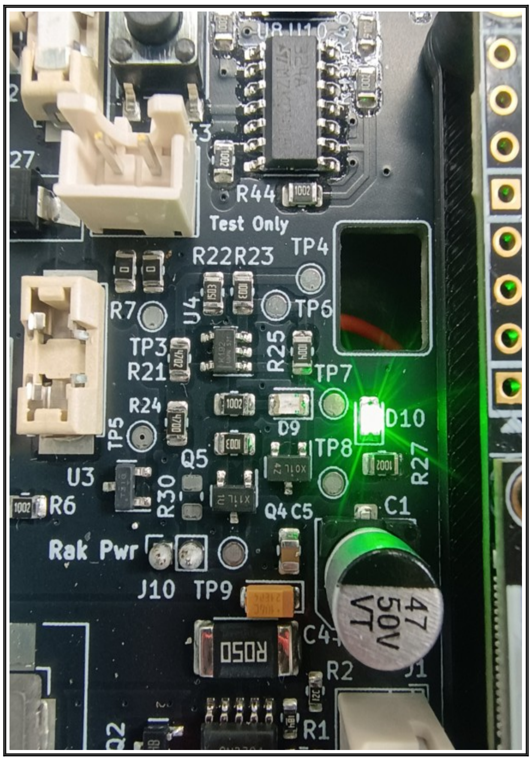
At times (particularly when the Solar panel is in bright sunlight) you may see both the Green (D1) and Red (D2) LEDs illuminated, but toggling on and off rapidly. This indicates that the Lithium-Ion batteries are almost at maximum charge (around 4.2v) and the MPPT charge controller is in constant current charging mode.

Over-discharge Protection Mechanism

It is important to remember there is only so much a MPPT controller can do to keep Lithium-Ion batteries charged if the just isn’t enough light. This is especially true if the software running on the RAK nRF52840 processor is very busy handling lots of LoRa packets (which can happen with MeshCORE and Meshtastic if the node is not at the edge of the mesh and is routing lots of MeshCORE or Meshtastic packets for other repeaters). As described above, in this scenario the most important thing is to prevent the Lithium-Ion batteries from being discharged too much, as this will effect the life span of the batteries. Over-discharging will have catastrophic consequences for a Lithium-Ion battery, especially large current over-discharge or repeated discharge, as this will damage the batteries permanently. To prevent this the MPPT charger PCB in you MeshCORE/Meshtastic repeater comes with a Lithium-Ion over-discharge protection circuit which prevents the Lithium-Ion batteries from dropping below 3.1V. If during a long period of dull or overcast weather (i.e. a period of no direct sunlight), the Solar Panel and MPPT controller cannot keep the batteries topped up (so the battery voltage drops to 3.1V), the over-discharge protection circuit will automatically kick in and turn off power to the RAKwireless WishMesh 1000mW (+30dBm) board (to prevent further discharge and damage). When this occurs, the over-discharge protection circuit will then work in conjunction with the MPPT controller to charge the Lithium-Ion batteries back up as quickly as possible, and, crucially, will only turn the RAKwireless WishMesh 1000mW (+30dBm) board back on when the Lithium-Ion cells reaches 3.7V. This hysteresis is deliberate to allow the Lithium-Ion cells to recover and be charged back up to their plateau voltage level (3.7v) which will allow your MeshCORE/Meshtastic repeater to come back up and stay up.

In order to indicate what the discharge protection circuit is doing, the MPPT controller PCB has two additional LEDs – D10 (Green) and D9 (Red) – which are positioned in the middle of the board, above the 47uF Electrolytic capacitor labelled C1. When the Green LED is illuminated (as shown in the picture below right), this indicates that the battery voltage is currently above 3.1v and all is well:





When the Red LED is illuminated (as shown in the picture above left), this indicates that the Lithium-Ion battery voltage (of the six combined 18650 cells) is currently below 3.1v and all is NOT well. If you use 18650 Lithium-Ion batteries in you RAKwireless WishMesh 1000mW (+30dBm) MeshCORE/Meshtastic repeater which have not been charged up and balanced (so they are the same voltage), it is possible everything will remain off if the discharge protection circuit detects that the combined voltage of all six 18650 batteries (as they are all in parallel) are below 3.1V. Therefore, always use new 18650 Lithium-Ion batteries, from a known manufacturer, that have been purchased at the same time; and make sure they are externally charged to the maximum (4.2v) before installing them.

When you purchased your RAKwireless WishMesh 1000mW (+30dBm) MeshCORE/Meshtastic repeater, if you did not purchase 18650 batteries (or you are located outside the UK where sadly we cannot supply Lithium-Ion batteries), please ensure you follow the above advice when purchasing your own. I recommend either EVE 33V 3200mAH or EVE 33VA 3300mAH flat-top batteries from battery101.co.uk (UK) or nkon.nl (EU) or 18650batterystore.com (US).

UK

EU

US

Please scan the QR codes above using you phone to visit these online shops

18650 Lithium-Ion Batteries and Fuses

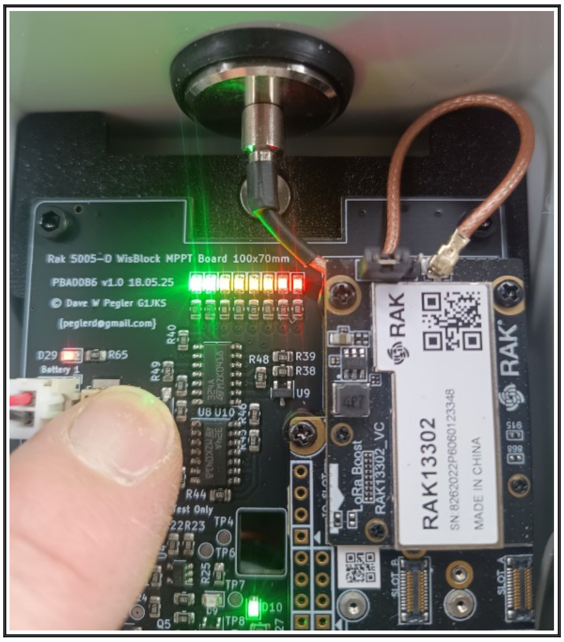
In addition to the over-discharge protection circuit, the battery PCB in your RAKwireless WishMesh 1000mW (+30dBm) MeshCORE/Meshtastic repeater (located in the door of the IP67 case) has six 1808 1A protection fuses. These fuses are provided in case you accidentally insert the 18650 batteries the wrong way around, or if one of the 18650 cells fails and goes short-circuit.

The six 18650 Lithium-Ion cells in the door of your RAKwireless WishMesh 1000mW (+30dBm) Meshtsatic/MeshCORE repeater are connected in parallel, so if one of the cells fails and goes short circuit, this could shorten out the other cells causing a large current to flow. This could cause overheating, so the 1A 1808 protection fuses are provided to blow should this ever occur.

Note that the 18650 Lithium-Ion cells need to be inserted correctly, with the **negative** and **positive** terminals oriented so that positive is pointing towards the 2.54mm (red and black) power connector as show in the picture above. To help you, there are **+** and **–** symbols marked on the 18650 battery holders. If you accidentally blow the 1808A fuses (by connecting one of the batteries the wrong way around) then don't fret, they are there in case this happens. You can easily obtain new fuses from Amazon, eBay, Ali express or any good electronics retailer. They are very cheap, just search for "1808 1A Fuses".

18650 Lithium-Ion Battery Tester

To quickly check the 18650 Lithium-Ion battery charge level, at the top of the MPPT controller PCB is a simple eight LED bar graph battery tester, which can be initiated by pressing the switch situated next to the JST battery connector labelled "Battery 1":



The battery tester consists of a bar graph of eight coloured LEDs (two RED, four YELLOW and two GREEN) which are used to indicate what the battery charge level is. If the batteries are fully charged, all eight LEDS will come one when the battery test switch is pressed (as shown above). This indicates that the combined battery voltage is over 4.0V. If 7 LEDs come on (so just one GREEN LED), this indicates that the battery level is approximately 3.7V. If no GREEN LEDs come on, then this indicates the battery voltage is around 3.5V, and if only the RED LEDs come on, then this indicates that the battery voltage is probably below the over-discharge protection threshold of 3.1V (and D9 will be illuminated to indicate this).

Installation Overview

With all things radio, height, the type of antenna, and minimising cable insertion loss are key to getting good range – both for transmit and receive. Although LoRa is great for Long Range, low power communications (it uses something called Chirp Spread Spectrum to achieve this), even it is useless if the antenna is poor, the equipment being used is at ground level (where there are lots of things like building in the way to attenuate the signal), or if a substantial proportion of the RF signal is lost between the equipment generating the signal (e.g. the RAKwireless WishMesh 1000mW board) and the antenna. This is particularly true at the UHF frequencies that Meshtastic and MeshCORE uses (i.e. 866MHz).

The best way to overcome these losses and hence maximise range, is to mount the electronics generating (and receiving) the RF signal as close to the antenna as possible (so that losses in any RF cables are as small as possible), use a antenna with some gain, and mount the complete assembly as high as possible. For this reason the RAKwireless WishMesh 1000mW (+30dBm) MeshCORE/Mehtastic 25W solar repeater you have purchased has been designed so that the electronics generating (and receiving) the RF signal are co-located with a high gain antenna, all in the same IP67 enclosure. The electronics are contained inside the IP67 enclosure, and the antenna is mounted on the outside, with a short length of coaxial cable (10cm) between them. This way the RF feeder losses are kept to an absolute minimum, and the gain of the antenna extends the range. The complete unit (IP67 case, electronics and high gain antenna) can then be mounted in an elevated location using the external 1.5” (38mm) 3D printed mounting brackets – typically to a standard 38mm aluminium mast or satellite mounting bracket. The aluminium mast and brackets are not supplied with your RAKwireless WishMesh 1000mW (+30dBm) MeshCORE/Meshtastic repeater but they are available from many online retailers. Just make sure the diameter of the pole that you want to mount everything on is 38mm,.

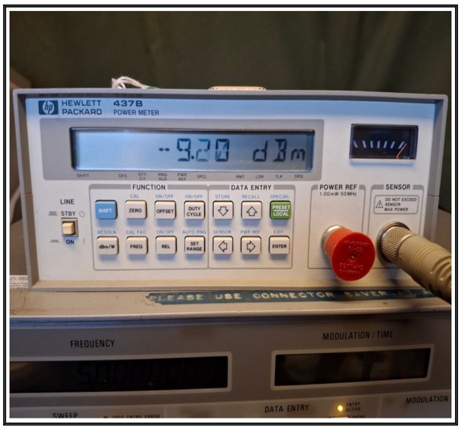
Before shipping your RAKwireless WishMesh 1000mW (+30dBm) 25W solar node, it is configured as a repeater and tested as such. We test that the SX1262 LoRa modem and SkyWorks SKY66122 PA are transmitting and that receive is working correctly. To do this we use the official MeshCORE flasher utility at **meshcore.io** (not meshcore.co.uk) to install the latest version of firmware, and then use the “Repeater Setup” utility to configure the repeater or room server settings.

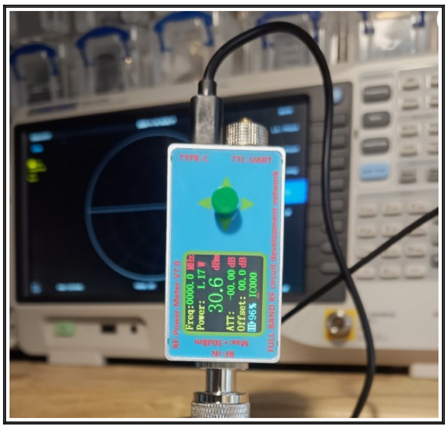
Below is a QR code which will take you to this utility at meshcore.io:

If you ever need to re-flash the firmware, make sure you select the “RAK WisMesh 1W Booster (3401 +13302) option as the target device and then following the instructions.

Warning: Legal Power Levels

When we install the firmware and test a RAKwireless WishMesh 1000mW (+30dBm) 25W solar repeater prior to shipping, we test that receive and transmit from the RAK 13302 FEM board (which contains the SX1262 LoRa modem and SkyWorks SKY66122 PA) is working correctly. This requires us to set the output power level to +30dBm (or 1W). When we do this we direct the RF energy through a RF power meter (e.g. Bird Thruline Model 43 with a 1000mW UHF 800-1000MHz slug, or HP 437B power meter) and then dump it into a 25W dummy load. We do not direct into an antenna.





To do the latter would be illegal in the UK (and EU), as the maximum permitted output power (as stipulated by Ofcom document IR2030) in the license exempt 868MHz ISM band is +27dBm (500mW) EIRP (Effective Isotropic Radiated Power); with a 10% duty cycle. The EIRP calculation includes the power of the transmitter, minus any transmission-line insertion loss (e.g. losses in the coax) plus any antenna gain.

To read more about permitted power levels in the UK, please use your phone to scan the following QR code, which will take you to the Ofcom (UK spectrum regulator) IR2030 documentation:

Alternatively, click on the following URL :

https://tinyurl.com/bdfcwp7b

It is my understanding that the +27dBm power limit (EIRP) is also in force in the EU, but not in the US. I therefore urge you to find out more and ensure the EIRP level is below the stipulated limit for your country. Although the RAKwireless WishMesh 1000mW (+30dBm) 25W Solar repeater is capable of transmitting above legal limits in many countries, it is your responsibility to ensure it does not. The manufacture of my car says it will do 120mph. Of course, I have no idea if this true as the maximum legal limit in the UK is 70mph, and it is my responsibility not to exceed this. The same is true for RF power levels.

When we ship we leave the power level in the meshcore.io repeater configuration utility set to +22dBm, which with the additional gain of the SkyWorks SKY66122 PA (+8dBm) results in an output power of +30dBm. If you plug in the +5.5dBi gain antenna supplied with the repeater, this will result in an EIRP of +35.5dBm (+30dBm + 5.5dBi) or 3.5 Watts – which is 7 times over the permitted limit in the UK/EU. Therefore for UK and EU customers I recommend setting the “TX power level” in the **meshcore.io** configuration utility to no more than +13.5dBm to ensure compliance (+13.5dBm+8dBm+5.5dBi = +27dBm EIRP) with IR2030.

73 Dave G1JKS